

Software Engineering Lab Project Report

On

AI-DRIVEN ENGLISH COMMUNICATION PLATFORM

Submitted in partial fulfillment of the requirements for the End Semester Exam of

SOFTWARE ENGINEERING LAB(20CS3183)

In

Third Year Semester-I

By

MODUKURU SRIDHAR (RO200075)

BADAM V PAVAN CHANDRA LOHITH (RO200135)

MUVVA SREE LASYA (RO200241)

PAGADALA ABHIRAM (RO200382)

PALLAKI PRAVALLIKA (RO200652)

Under the Guidance of

Dr. MSR Kiran Nag

Assistant Professor

Department of Computer Science and Engineering



RAJIV GANDHI UNIVERSITY OF KNOWLEDGE AND TECHNOLOGIES-A.P.

ONGOLE CAMPUS

Kurnool Road, ONGOLE, Prakasam Dt., Andhra Pradesh-523225

CERTIFICATE

This is to certify that the project entitled “ **AI-DRIVEN ENGLISH COMMUNICATION PLATFORM** ” being submitted by **MODUKURU SRIDHAR (RO200075) , BADAM V PAVAN CHANDRA LOHITH (RO200135) , MUVVA SREE LASYA (RO200241) , PAGADALA ABHIRAM (RO200382) and PALLAKI PRAVALLIKA (RO200652)** in partial fulfillment of the requirements for the Science and Engineering to **Rajiv Gandhi University of Knowledge Technologies - A.P. ONGOLE Campus**, is a bonafide work carried out by them under my guidance and supervision from **June 2024 to November 2024**.

The results presented in this project have been verified and found to be satisfactory. The results embodied in this project report have not been submitted to any other University for the award of any other degree or diploma.

Dr. MSR Kiran Nag
Assistant Professor,
CSE Department,
RGUKT, ONGOLE.

Mr.N.Mallikarjuna,
Head of the Department,
CSE Department,
RGUKT, ONGOLE.

APPROVAL SHEET

This report entitled “ **AI-DRIVEN ENGLISH COMMUNICATION PLATFROM** ” submitted by Modukuru Sridhar , Badam V Pavan Chandra Lohith , Muvva Sree Lasya , Pagadala Abhiram , Pallaki Pravallika is approved for the degree of Bachelor of Technology in COMPUTER SCIENCE AND

Examiner(s)

Supervisor

Date : _____

Place: _____

DECLARATION

We hereby declare that the project entitles “**AI-DRIVEN ENGLISH COMMUNICATION PLATFORM**” submitted to the **Rajiv Gandhi University of Knowledge Technologies - A.P. ONGOLE Campus** in partial fulfillment for the End Semester Lab Exams of Third Year Semester- I in Computer Science and Engineering is a record of an original work done by us under the guidance of **Dr MSR KIRAN NAG Assistant Professor** any university for the award of any other degree or diploma.

MODUKURU SRIDHAR (RO200075) _____

BADAM V PAVAN CHANDRA LOHITH (RO200135) _____

MUVVA SREE LASYA (RO200241) _____

PAGADALA ABHIRAM (RO200382) _____

PALLAKI PRAVALLIKA (RO200652) _____

Date:

Place:

ACKNOWLEDGEMENT

We convey our profound thanks and gratitude to our Director RGUKT ONGOLE CAMPUS **Dr . BHASKAR PATEL** for providing us an excellent academic encouragement and constant help. We extend our sincere thanks to our Academic Dean **Mr. MEESALA RUPAS KUMAR**, for his climate which made this endeavor possible.

We express our sincere gratitude to **Mr . MALLIKARJUN** HOD -CSE for his suggestions, motivations and co-operation for the successful completion of the work.

It is our privilege to express a profound sense of respect , gratitude and indebtedness to our guide **DR.MSR KIRAN NAG Assistant Professor** , Department of Computer Science and Engineering, Rajiv Gandhi University Of Knowledge Technologies – A.P. ONGOLE Campus,for her indefatigable Inspiration , guidance , cogent discussion , constructive criticisms and encouragement throughout the dissertation work.

MODUKURU SRIDHAR	RO200075
BADAM V PAVAN CHANDRA LOHITH	RO200135
MUVVA SREE LASYA	RO200241
PAGADALA ABHIRAM	RO200382
PALLAKI PRAVALLIKA	RO200652

ABSTRACT

With the increasing demand for English language proficiency across professional and academic domains, there is a need for tools that provide real-time, interactive learning experiences to improve communication skills. The AI-Driven English Communication Platform is designed to meet this need through three innovative features: (1) Group Discussion – facilitating simulated discussions among four participants, helping users develop conversational fluency in a collaborative setting; (2) One-on-One Conversations – enabling voice-to-text transcription and text-to-voice response, allowing users to engage in dynamic, AI-powered conversations to enhance both spoken and listening skills; and (3) Slide Summarizer – generating concise summaries of each slide in an uploaded PPT file, aiding comprehension and study efficiency.

The platform leverages a range of technologies to bring these features to life. The python-pptx library is used to process and extract information from PPT files, enabling the slide summarizer feature. Pillow handles image manipulation tasks, such as formatting and preparing slide visuals for processing. Google-generativeai powers the natural language generation, assisting with response generation in the conversation modules. Groq provides the computational speed and efficiency necessary for real-time responses, and deepgram-sdk is employed for accurate speech-to-text and text-to-speech conversion, ensuring a seamless conversational experience.

The model's performance is evaluated by measuring accuracy in transcription and response generation, as well as user satisfaction in interaction and comprehension tasks. Initial results demonstrate that the platform can effectively support English language learning, making it a valuable tool for improving communication skills.

CONTENT

S.NO	TITLE	PAGE NO
1.	INRTODUCTION	1
	1.1.MOTIVATION	
	1.2.PROBLEM DEFINITION	
	1.3.OBJECTIVES OF THE PROJECT	
2.	LITERATURE SURVEY	2
3.	ANALYSIS	3-4
	3.1.EXISTING SYSTEM	
	3.2.PROPOSED SYSTEM	
	3.3.SOFTWARE REQUIREMENT SPECIFICATION	
4.	DESIGN	5-11
	4.1.UML DiAGRAMS	
5.	IMPLEMENTATION	12-19
	5.1.MODULES	
	5.2.MODULE DESCRIPTION	
	5.3.INTRODUCTION OF TECHNOLOGIES USED	
	5.4.SAMPLE CODE	
6.	TEST CASES	20-21
	6.1BLACK BOX TESTING	
	6.2 WHITE BOX TESTING	
7.	OUTPUT SCREENSHOTS	22-25
8.	CONCLUSION	26
9.	FUTURE ENHANCEMENT	27
10.	BIBLIOGRAPHY	28

1.INTRODUCTION

In today's globalized world, effective English communication skills have become essential in both professional and academic settings. However, language learners often face challenges in improving their fluency, comprehension, and conversational skills, especially without a structured environment for practice. This AI-driven English communication platform aims to address these challenges by offering users an interactive, AI-powered tool to practice and enhance their English communication skills. The platform provides three main features: a Group Discussion module that allows collaborative conversation among four participants, a One-on-One Conversation feature with voice-to-text and text-to-voice responses, and a Slide Summarizer that generates summaries for uploaded PPT slides to aid comprehension.

1.1.Motivation:

With the increasing demand for English proficiency in diverse fields, learners need practical, interactive tools to improve their communication skills. This platform is designed to make language learning more accessible, engaging, and effective by integrating advanced AI technology for real-time feedback and assistance. By fostering an immersive language practice environment, this project aims to help users build confidence in their spoken and written English skills, contributing to their personal and professional growth.

1.2.Problem Definition:

The lack of accessible, interactive platforms for English language practice limits learners' ability to develop fluency and comprehension, while traditional methods lack real-time feedback and content summarization tools, hindering effective language skill development and information retention.

1.3.Objective of the project:

The objective of this project is to develop an AI-driven platform that enhances English communication skills through immersive, interactive tools. The platform provides a Group Discussion feature for practicing conversational fluency in a collaborative setting, a One-on-One Conversation module using voice-to-text and text-to-voice responses for real-time conversational feedback, and a Slide Summarizer that generates summaries of uploaded PPT slides to aid comprehension. By integrating NLP and speech processing tools like python-pptx, Pillow, google-generativeai, Groq, and deepgram-sdk, the project aims to offer a practical solution that supports language learners in building confidence, fluency, and comprehension skills effectively.

2.LITERATURE SURVEY

Here's a literature survey for your AI-driven English communication platform project:

Benefits:

Extensive studies in AI and language learning indicate that interactive, AI-powered tools can significantly improve users' language skills by offering real-time feedback and immersive practice. Integrating natural language processing (NLP) and speech recognition technologies enables personalized language practice, helping users enhance fluency, comprehension, and conversational skills. AI-driven platforms also provide tailored learning experiences, allowing learners to practice in a way that adapts to their individual progress. Research supports that tools like group discussions and conversational AI can create engaging learning environments, which are crucial for improving language retention and confidence.

Challenges:

Building an AI-driven communication platform poses several challenges. Ensuring accurate speech recognition and natural response generation is crucial, especially in a diverse user base with varying accents and speech patterns. Additionally, generating coherent and relevant summaries for slide presentations requires sophisticated NLP techniques to extract key points without omitting important context. Another challenge is maintaining a user-friendly interface that integrates these complex technologies in a way that remains accessible and intuitive for users, as complex interfaces may overwhelm language learners.

Solution:

The AI-driven English communication platform leverages advanced NLP algorithms and speech recognition tools for accurate transcription and response generation. We use python-pptx and Pillow for handling slide content, while google-generativeai and deepgram-sdk enable effective conversation modules. To ensure real-time processing and seamless user experience, Groq provides high-performance computing capabilities. The platform also employs design best practices to create a user-friendly interface that clearly presents tools and insights, making language practice efficient and engaging.

3.ANALYSIS

3.1.EXISTED SYSTEM

Existing English language learning tools and platforms generally fall into two categories: static content-based methods and basic conversational applications. Content-based methods, such as language learning websites and mobile apps, provide structured lessons or vocabulary-building exercises, but they lack real-time interaction and conversation practice. Basic conversational applications offer limited speech-to-text and text-to-speech functionalities but often lack advanced natural language processing (NLP) capabilities to handle complex language interactions or provide meaningful feedback. Additionally, most existing platforms do not include tools for summarizing or processing visual information, such as PowerPoint slides, which limits their application for learners seeking comprehensive comprehension skills.

3.2.PROPOSED SYSTEM

Our proposed system, the AI-Driven English Communication Platform, includes several innovative features designed to enhance language learning in an interactive, real-time environment. These new functionalities are:

Group Discussion Module – This feature enables up to four participants to engage in structured discussions, simulating real-world conversations to build fluency, confidence, and group communication skills.

One-on-One Conversation Module – Users can interact with AI in a simulated conversation by speaking into the platform, which converts their voice input into text and generates responses in both text and voice formats. This module helps users practice listening and speaking skills through real-time feedback.

Slide Summarizer – This tool allows users to upload PowerPoint (PPT) files, and the system generates a concise summary of each slide. This feature aids users in quickly comprehending key points, which is particularly useful for academic or business presentations.

Dynamic User Dashboard – The dashboard is designed for ease of use, allowing users to navigate seamlessly between modules. Users can upload content, initiate conversations, join discussions, and access summaries, all within a single, streamlined interface.

Real-time Language Processing Tools – Our system integrates advanced NLP and speech processing technologies, using python-pptx for slide processing, Pillow for image handling, google-generativeai for language generation, Groq for performance optimization, and deepgram-sdk for accurate speech recognition.

This enhanced platform offers an all-in-one solution for English communication practice, combining group interaction, real-time conversation, and content comprehension tools to provide users with a comprehensive language learning experience.

3.3.SOFTWARE REQUIREMENT SPECIFICATION

Our project contains front end and Back end . In the process of creation of website we use different programming languages in the proposed system.

Front end Programming Languages:

- HTML
- CSS
- JavaScript

Back end Programming Languages:

- Python

Hardware Requirements:

- Laptop
- Internet Connection

By using the Above mentioned Software and Hardware requirements we completed our project much successfully.

3.3.1.PURPOSE:

The purpose of this AI-driven English communication platform is to provide users with a real-time, interactive language learning environment. The platform enables users to participate in group discussions engage in one-on-one conversations with AI-driven responses, and use a slide summarizer for understanding key points in presentations. By incorporating AI-powered speech-to-text, text-to-speech, and summarization technologies, the platform helps users develop fluency, confidence, and comprehension in English. Each feature is designed to deliver real-time feedback, making language learning practical and engaging

3.3.2.SCOPE:

The platform is designed to dynamically process user interactions, including voice and text input, and generate accurate, context-aware responses. It allows users to practice speaking, listening, and comprehension skills in one environment. Additionally, the slide summarizer offers detailed yet concise summaries of uploaded presentations, supporting academic and business users who require efficient learning or review tools.

3.3.3.OVERALL DESCRIPTION:

The AI-driven English communication platform is an interactive tool aimed at improving English language skills. It features group discussions for collaborative practice, one-on-one AI-assisted conversations for real-time feedback, and a slide summarizer for quick comprehension of presentation content. Speech recognition, language processing, and summarization tools, including python-pptx, Pillow, google-generativeai, Groq, and deepgram-sdk, provide accurate and seamless interaction. The platform visually presents feedback and progress tracking, enabling users to monitor their development and identify areas for improvement effectively. This project ultimately aids users in becoming more proficient and confident English communicators.

4.DESIGN

4.1 UML DIAGRAMS:

U M L is the short form of Unified Modelling Language . A UML is a standardized general - purpose modeling language in the field of the object - oriented software engineering . The standard is managed, and was created by, the Object Management Group.The important goal for U M L is to create a common modelling language for the sake of Object Oriented Software engineering . In its current form of UML consists of two major components : a Meta - model and a notation.In the future, some form of method or process may also be added to; or associated with, UML.

Unified Modeling Language(UML) is a standard language for specifying,Visualization Constructing and documenting the artifacts of software systems, as well as for business modeling and other non-software systems.UML represents a collection of the best engineering practices that have proven successful in the modeling of large and complex systems.

UML is a very important part of developing object oriented software and software development process. The UML uses mostly graphical notationsto express the design of software projects.

GOALS

The Primary goals in the design of the UML are as follows:

1. Provide users a ready - to – use , expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development processes.
4. Provide a formal basis for understanding the modeling language
5. Encourage the growth of the OO tools market.
6. Support the higher level development concepts like such as collaborations frameworks, patterns and components and Integrate best practices

UML Diagram is based on UML(Unified Modeling Language) with the purpose of visually representing a system along with its main actors ,roles ,actions ,artifacts or classes,in order to better understand,alter, maintain,or for document information about the system. The UML diagrams are divided into **Structural** and **Behavioural** UML Diagrams.

4.1.1 STRUCTURAL UML DIAGRAMS:

Structural diagrams are a static view of the structure of a system. It is widely used in the Documentation of software architecture. The Structural UML Diagrams involves 2 diagrams They are:

- Class Diagram
- Component Diagram

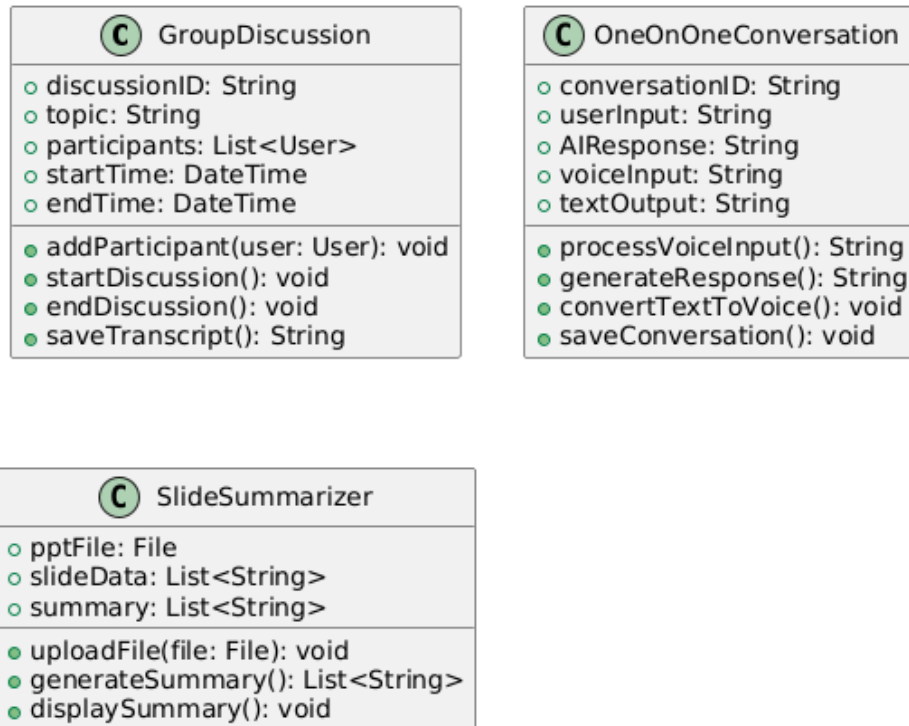
4.1.2 BEHAVIOURAL UML DIAGRAM:

Behavioural diagrams potrays a dynamic vew of a system or the behaviour of a system ,which describes the functioning the system. It involves 3 diagrams They are:

- Usecase Diagram
- Sequence Diagram
- Activity Diagram

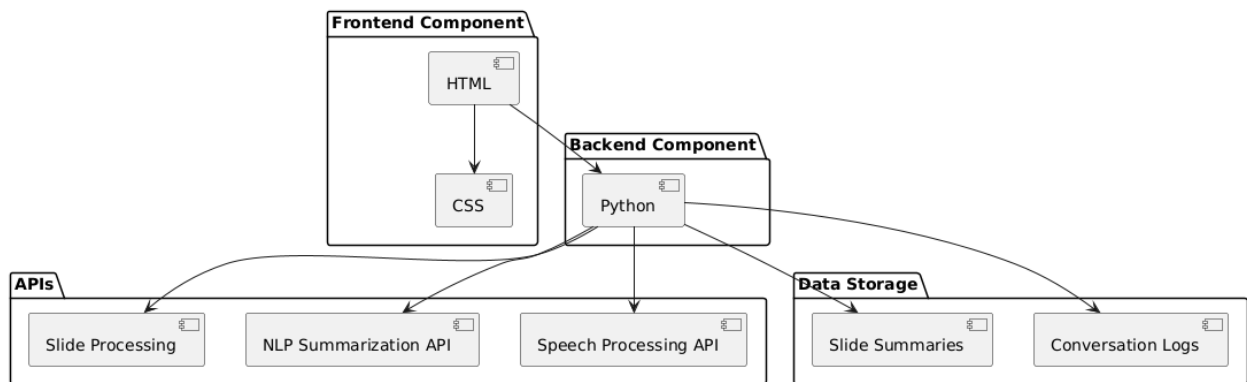
CLASS DIAGRAM

Class diagrams are one of the most widely used diagrams. It is the backbone of all the object-oriented software systems. It depicts the static structure of the system. It displays the system's class, attributes, and methods. It is helpful in recognizing the relation between different objects as well as classes.



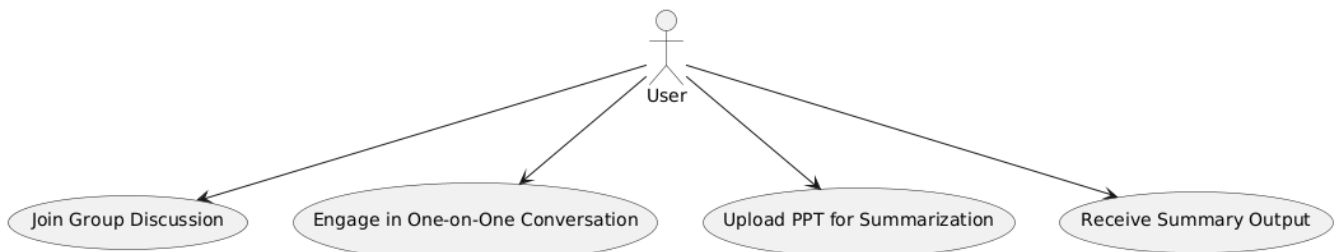
COMPONENT DIAGRAM

It portrays the organization of the physical components within the system. It is used for the modeling execution details. It determines whether the desired functional requirements have been considered by the planned development or not, as it has structural relationships between the elements of a software system.



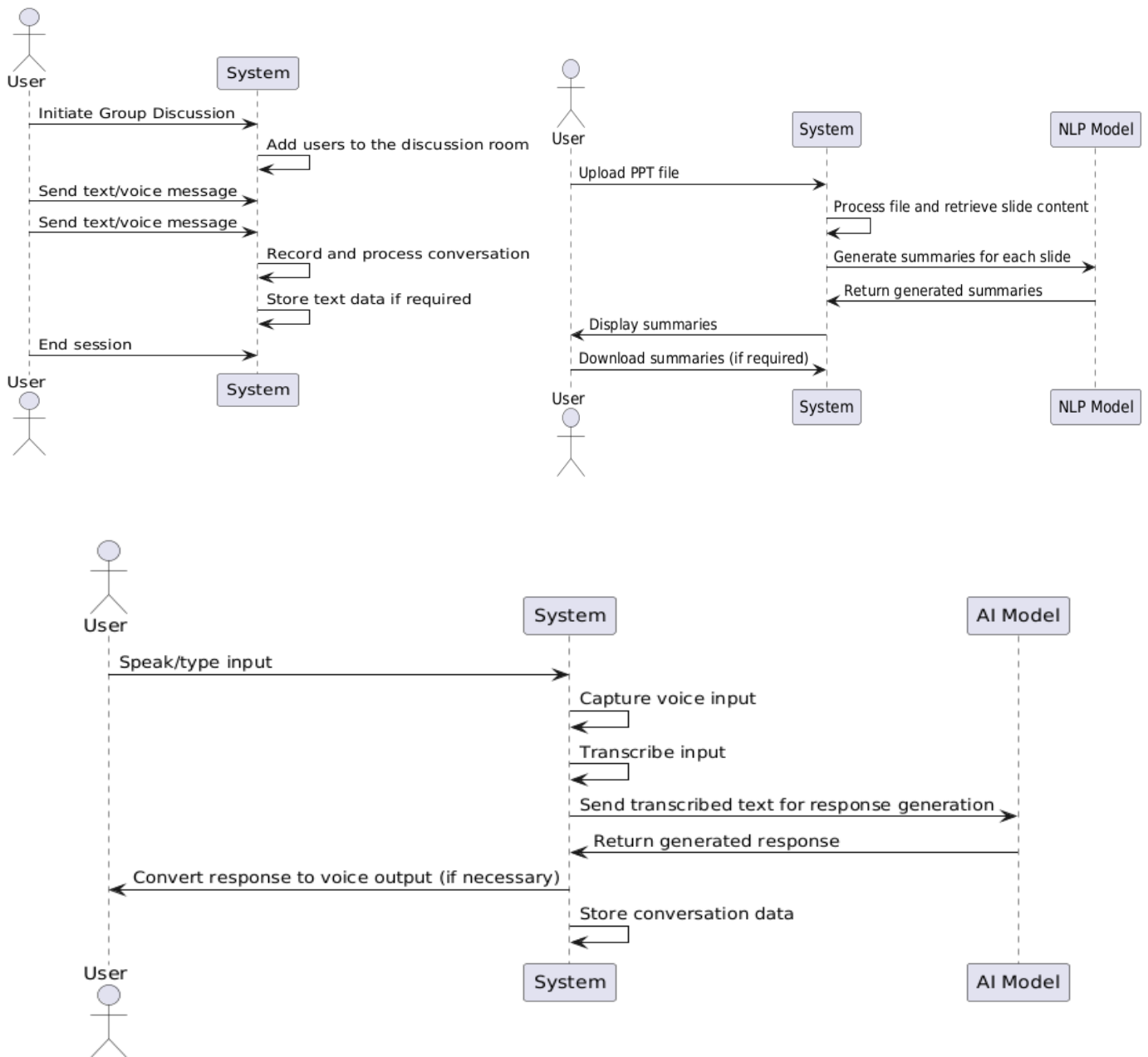
USECASE DIAGRAM

In U M L , use - case diagrams model the behavior of the system and help to capture the requirements of the system . Use-case diagrams describe the high-level functions and scope of a system. These diagrams also identify the interactions between the system and its actors . The use cases and actors in use -case diagrams describe what the system does and how the actors use it, but not how the system operates internally.



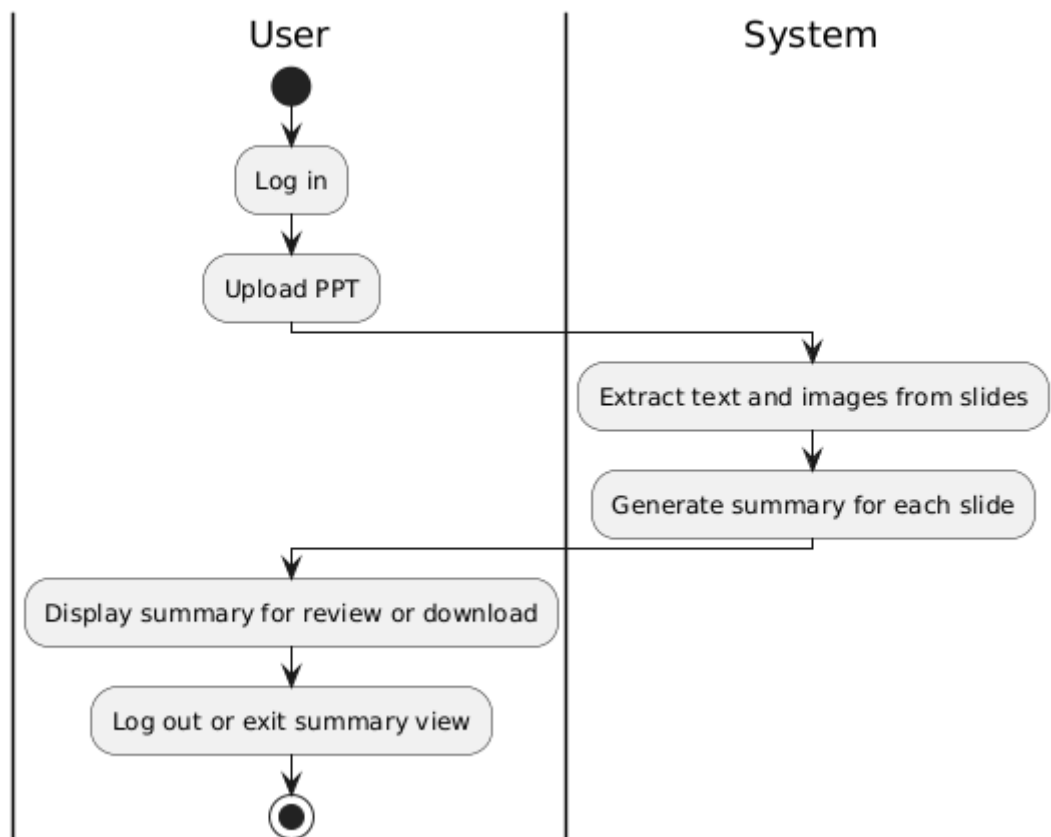
SEQUENCE DIAGRAM

The sequence diagram represents the flow of messages in system and is also termed as an event diagram. It helps in the envisioning several dynamic scenarios. It portrays the communication between of any two lifelines as a time-ordered sequence of events, such that these lifelines took part at the run time. In UML the lifeline is represented by a vertical bar , where as the message flow is represented by a vertical dotted line that extends across the bottom of the page. It incorporates the iterations as well as branching .



ACTIVITY DIAGRAM

In the UML, the activity diagram is used for demonstrate the flow of control with in the system rather than the implementation . It models at the concurrent and the sequential activities . The activity diagram helps in envisioning the workflow from one activity to the another. It puts emphasis on the condition of Flow and the order in which a activity occurs. The flow can be sequential, branched,or concurrent, and to deal with such kinds of flows , the activity diagram has come up with a fork , join , etc . It is also termed as an object-oriented flowchart . It encompasses activities composed of a set of actions or operations that are applied to model the behavioral diagram.



5. IMPLEMENTATION

5.1 MODULES:

- **Home page module**
- **User module**
- **Admin module**

5.2 MODULE DESCRIPTION:

Home Page module:

The home page serves as the main interface for users, featuring three interactive cards representing the key functionalities: Group Discussions, One-on-One Conversations, and Slide Summarization. Each card is designed for easy navigation; when a user clicks on a card, it redirects them to the corresponding feature's output page. The home page manages user interactions by providing clear pathways to access each module, ensuring a seamless experience. The layout is intuitive, promoting engagement and encouraging users to explore the platform's capabilities.

User module:

This module enables user interactions with the platform's features. Users can initiate group discussions, engage in one-on-one conversations, and upload PPT files for summarization. The module processes user inputs and presents results in an organized manner, facilitating easy access to the platform's core functionalities. Additionally, it handles conversation history and user feedback, providing an engaging and responsive experience.

Admin Module:

The Admin Module allows administrators to manage user data, monitor activity, and maintain platform integrity. It provides tools for user registration, access control, and system management, ensuring that the application runs smoothly and securely. This module also facilitates reporting features that allow admins to track usage patterns and user feedback, contributing to ongoing improvements in the platform's features and functionalities.

Admin module:

The Admin Module of the AI-driven English Communication Platform is a vital component that provides administrators with comprehensive access and control over the system. This module empowers administrators to efficiently monitor and manage user interactions, ensuring a smooth and secure user experience.

1. User Management Functions:

This feature enables administrators to add, modify, or remove user accounts. It allows for the monitoring of user activity and engagement levels, helping maintain the integrity of the platform and ensuring that users adhere to community guidelines.

2. Session Management :

Administrators can oversee ongoing group discussions and one-on-one conversations. This includes the ability to terminate sessions if necessary and review conversation transcripts for quality control and feedback purposes.

3. Data Analytics:

This function utilizes analytical tools to assess user engagement and feedback trends. By analyzing patterns in user interactions, administrators can derive valuable insights to inform future enhancements to the platform.

4. Error Handling :

The module incorporates robust error handling mechanisms using try and except blocks to manage exceptions that may arise during user management and session handling operations. This ensures a seamless experience by providing appropriate feedback and resolutions for any encountered issues.

5. Additional Information Pages :

About Page: This page outlines the purpose and mission of the AI-driven English Communication Platform, detailing its objectives and the benefits it offers to users.

Technical Details Page: This page provides insights into the technologies, libraries, and frameworks used in the development of the application.

Contact Form: This feature allows users to submit feedback or inquiries, promoting engagement and facilitating support for any issues or questions that may arise.

5.3 INTRODUCTION OF TECHNOLOGIES USED:

FRONT END:

HTML

Html or hypertext mark-up language is the main mark-up language for creating web pages and the other information that can displayed in a web browser . Html allows images and objects to be embedded and can be used to create interactive forms . It provides a means to create structural documents by denoting structural semantics for text such as headings, paragraphs, lists, links, codes and the other items . It can be embedded scripts written in the languages such as JavaScript which affects behaviours of html , web pages.

JavaScript:

JavaScript is a high-level, dynamic, and interpreted programming language primarily used for enhancing web pages and creating interactive web applications. Originally developed by Netscape, it has become a key technology in web development alongside HTML and CSS.

CSS:

Cascading style sheets (CSS) is style sheet language used for describing the look and formatting of a document written in a mark-up language . CSS specifies a priority scheme to determine in which style rules apply if more than one rule matches against particular element. In the so called cascade,priorities or weights are calculated and assigned to rules, so that the results are predictable.

BACKEND:

Python:

Python is a high-level, interpreted programming language known for its clear syntax and readability. Developed by Guido van Rossum and first released in 1991, Python has become one of the most widely used programming languages in the world. It is often praised for its versatility and ease of learning, making it a popular choice among beginners and experienced developers alike.

5.4 SAMPLE CODE:

App.py

```
from flask import Flask, render_template, request, jsonify, session
from components.responce_gen import generate_response
from components.resp_gen import generate_response_

app = Flask(__name__)
app.secret_key = 'your_secret_key' # Set a secret key for session management

chat_history = [

    for example:
    Ben : hello everyone, our topic today is Global warming and my thoughts on global warming are ...
    Tony: I agree with you Ben. We can reduce it by taking preventive measures.
    Steeve: My opinion is to plant trees...

    end this here, just one chance for each.
    most importantly everyone must definitely
    now the topic is:{}
    """"}
]
@app.route('/')
def home():
    return render_template('home2.html')

@app.route('/group')
def index():
    return render_template('app_index.html')

@app.route('/group/start_discussion', methods=['POST'])
def start_discussion():{"role": "system", "content": """"You are a Discussion generator. Suppose if the topic
is Global warming.
Three participants, Ben, Tony, and Steeve, will discuss the topic in a structured way.
just one exchange each

topic = request.form['topic']
chat_history[0]["content"] = chat_history[0]["content"].format(topic)

response_text = generate_response('gsk_hdgKKj3mXIXV3eZBW8VfWGdyb3FYfVzlfKlpghqgGV
VBZCF2DnV4', chat_history)

session['ben_output'] = response_text['ben']
session['tony_output'] = response_text['tony']
session['steeve_output'] = response_text['steeve']

return jsonify({
    'ben_output': session['ben_output'],
    'tony_output': session['tony_output'],
    'steeve_output': session['steeve_output']
})
```

```

@app.route('/group/get_response', methods=['POST'])
def get_response():

    user_input_ = request.form['user_input']

    ben_output = session.get('ben_output')
    tony_output = session.get('tony_output')
    steeve_output = session.get('steeve_output')

    suggestion_dict = [
        {'role': 'system', 'content': 'you are a vocabulary and grammar expert who upgrades the user inputs without changing or opposing that,just upgrade the sentence in terms of sentence formation.and most importantly give the response in form of html structure that can be placed inside a div, Example: user: playing games is a good thing as it is good for health. response:<b>in my opinion playing is really good for our physical health and there are also a lot of benifits.</b> example2: user: Global warming is good for earth suggestion: in my opinion, global warming is the Greatest thing that can happend to earth.'},
        {'role': 'assistant', 'content': f"ben: {ben_output} tony: {tony_output} steeve: {steeve_output}"},
        {'role': 'user', 'content': f"user: {user_input_}" }
    ]

    suggestion = generate_response_('gsk_hdgKKj3mXIXV3eZBW8VfWGdyb3FYfVzlfKlpghqgGVVBZC
F2DnV4', suggestion_dict)

    new_chat_history = [
        {
            "role": "system",
            "content": f""""You are continuing a group discussion based on the following responses,responces must not exceed 3-4 sentences for each.:
Ben: {ben_output}
Tony: {tony_output}
Steeve: {steeve_output}
User: {user_input_}
Continue the discussion with one response each for Ben, Tony, and Steeve."""""
        }
    ]

    response_text = generate_response_('gsk_hdgKKj3mXIXV3eZBW8VfWGdyb3FYfVzlfKlpghqgGVVBZ
CF2DnV4', new_chat_history)

    session['ben_output'] = response_text['ben']
    session['tony_output'] = response_text['tony']
    session['steeve_output'] = response_text['steeve']

    return jsonify({
        'ben_output': session['ben_output'],
        'tony_output': session['tony_output'],
        'steeve_output': session['steeve_output'],
        'user_suggestion': suggestion
    })

```

```

from flask import Flask, render_template, request, jsonify
# from pipeline import run_pipeline
from flask_cors import CORS
import os
import asyncio

from components.transcription import transcribe_audio
from components.respo_gen import generate_response___
from components.tts import text_to_speech

chat_history_ = [
    {"role": "system", "content": """" You are an English Communication voice chatbot called Peter.
    You have to reply very short(just in two or three sentences) and sweet. """"},
    {"role": "assistant", "content": "Hello, I am Peter. I'll help you learn English. What shall we start with? Is
    there any topic you'd like to talk or learn about?"}
]

# app = Flask(__name__)
CORS(app)

UPLOAD_FOLDER = 'uploads'
OUTPUT_FILE = 'static/output.wav'

if not os.path.exists(UPLOAD_FOLDER):
    os.makedirs(UPLOAD_FOLDER)

@app.route('/chat')
def indexx():
    return render_template('index.html')

@app.route('/chat/upload', methods=['POST'])
async def upload_file():
    if 'file' not in request.files:
        return jsonify({'error': 'No file part'}), 400

    file = request.files['file']
    if file.filename == "":
        return jsonify({'error': 'No selected file'}), 400

    filepath = os.path.join(UPLOAD_FOLDER, 'recording.wav')
    file.save(filepath)

    # Transcribe the audio asynchronously
    user_input = await asyncio.to_thread(transcribe_audio, 'ddb32fde2fa328ab0f486c251de09ce1492be6d',
        filepath)

    if not user_input.strip(): # Check if transcription is empty
        return jsonify({'error': 'No voice detected'}), 400

    print(user_input)
    chat_history_.append({"role": "user", "content": user_input})

```

```

response_text = await asyncio.to_thread(generate_response____, 'gsk_hdGKKj3mXIXV3eZBW8VfWG
dyb3FYfVzlfKlpghqgGVVBZCF2DnV4', chat_history_)
print(response_text)

output_file = os.path.join('static', 'output.wav')
await text_to_speech('fa514b70902b8a222eadfacf4920d2e0252be990', response_text, output_file)

chat_history_.append({"role": "assistant", "content": response_text})

# Return the transcription and the audio URL
return jsonify({
    'message': user_input,
    'response': response_text,
    'audio_url': f'/static/output.wav'
}), 200
# =====
import os
from flask import redirect, url_for
from werkzeug.utils import secure_filename
from components.extract_images import extract_text_and_images_from_pptx
from components.generate_content import process_images

UPLOAD_FOLDER__ = 'uploaded_files'
IMAGE_FOLDER__ = 'slide_images'
ALLOWED_EXTENSIONS = {'pptx'}

# app = Flask(__name__)
app.config['UPLOAD_FOLDER'] = UPLOAD_FOLDER__
app.config['IMAGE_FOLDER'] = IMAGE_FOLDER__
app.config['API_KEY'] = "AIzaSyC88AWwQQiL8_ZYWkmflut5MnxkfiOYqt4"
app.config['MODEL_NAME'] = "gemini-1.5-flash"

os.makedirs(UPLOAD_FOLDER__, exist_ok=True)
os.makedirs(IMAGE_FOLDER__, exist_ok=True)

def allowed_file(filename):
    return '.' in filename and filename.rsplit('.', 1)[1].lower() in ALLOWED_EXTENSIONS

@app.route('/slide')
def indexes():
    return render_template('slide_index.html')

@app.route('/slide/upload', methods=['POST'])
def upload_file():
    if 'file' not in request.files:
        return redirect(url_for('indexes'))

    file = request.files['file']

    if file and allowed_file(file.filename):
        filename = secure_filename(file.filename)
        file_path = os.path.join(app.config['UPLOAD_FOLDER'], filename)
        file.save(file_path)

```

```

# Extract images and text from the PPTX file
extracted_data = extract_text_and_images_from_pptx(file_path, app.config['IMAGE_FOLDER'])

# Prepare the prompt for the Gemini model
prompt_template = """You are smart ai assistant that generates the elobarates content for giving a
presentation by providing images and the text present in the slide, elobarate the
content of the slide in such a way that it is in a specific manner.for example: if the
text in the slide is: Rose Hibiscus Lily and the images are of these flowers, you
should explain like
1.take a this beautiful flower: Rose, it is red in color and the most beautiful flower.
2.and the next one is hibisus is ....
3... in this way you how to think about the pictures and text, relate them and form
a continuous flow of presentation.no need of introduction as you are in the middle
of presentation unless the text is asking for it. Give the responce in continuous
normal text without any effects, as new line or special characters might result in
** and /n now the text is : {text}
"""

results = []
count = 0
for slide_data in extracted_data:
    text=slide_data['text']
    count += 1
    if count == 1:
        text = 'This is first slide, start by wishing and introducing the topic and slowly get into this
        Text:' + text
    if count == len(extracted_data):
        text = 'Finally'+text+'thank you.'

    prompt = prompt_template.format(text=text)
    images_folder = slide_data['images']
    slide_results = process_images(app.config['API_KEY'], app.config['MODEL_NAME'],
                                images_folder, prompt)

    Print("=====")
    print(slide_results)
    Print("=====")
    results.append(slide_results)

return render_template('result.html', results=results)

return redirect(url_for('indexes'))

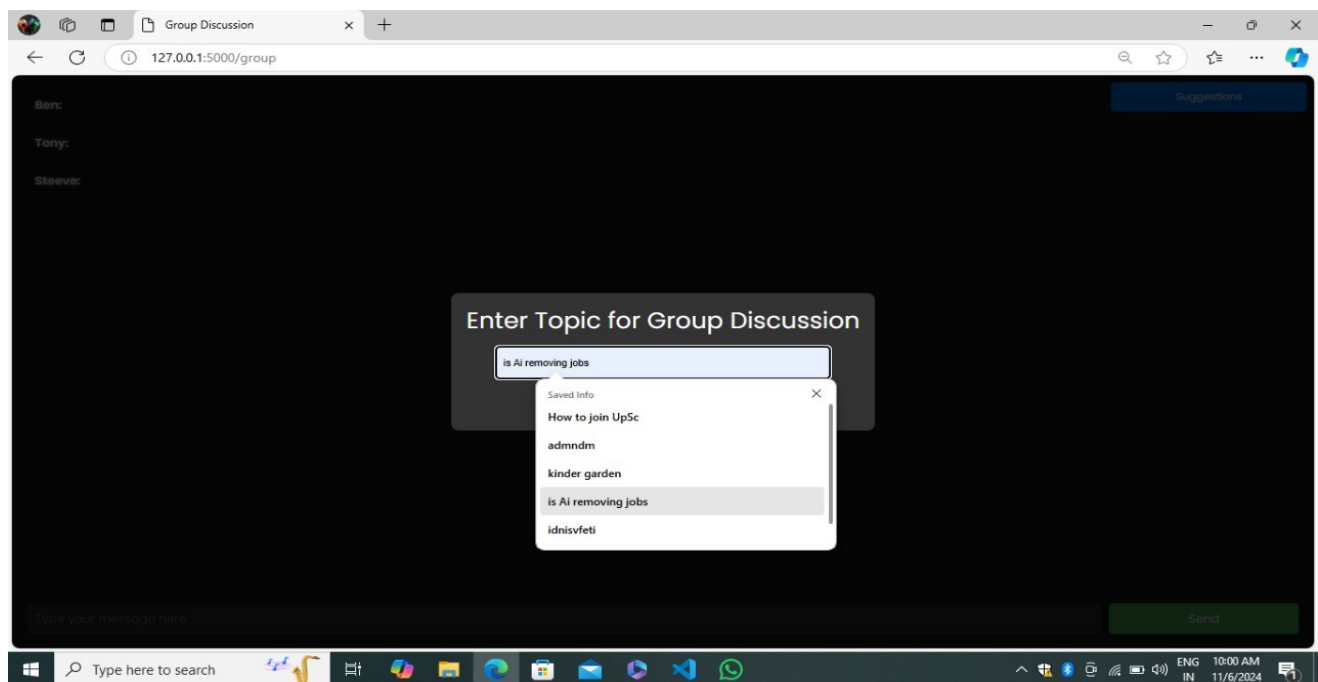
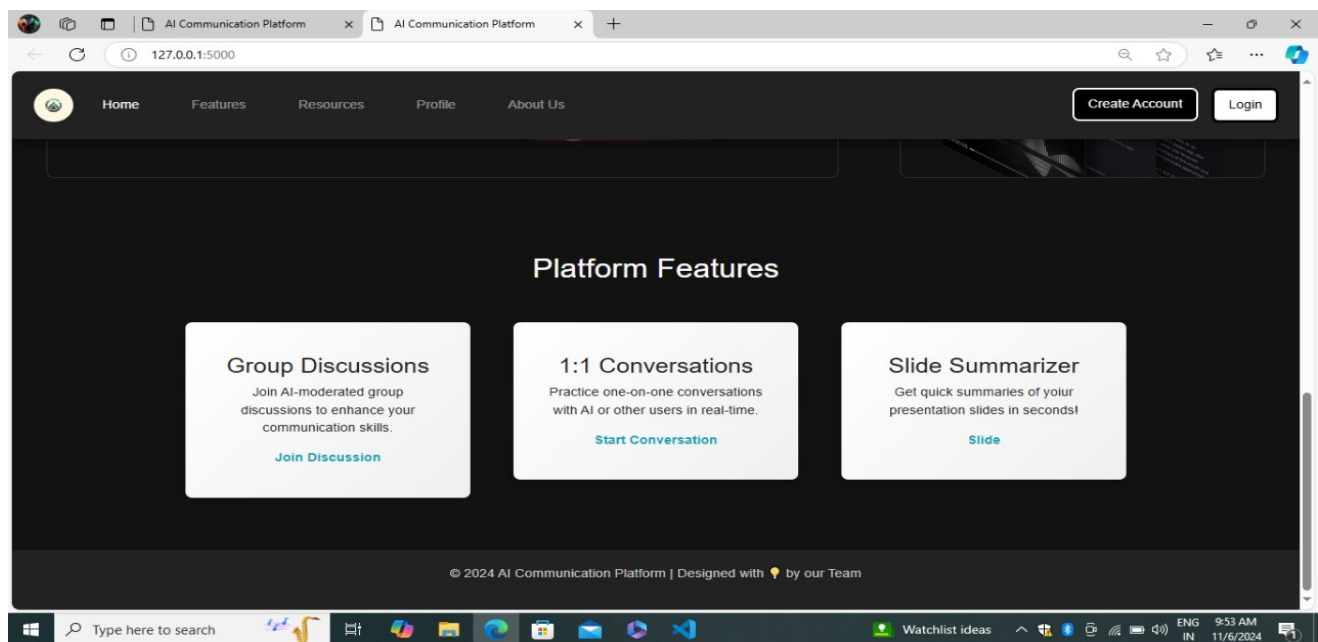
if __name__ == '__main__':
    app.run(debug=True)

```

6.TEST CASES

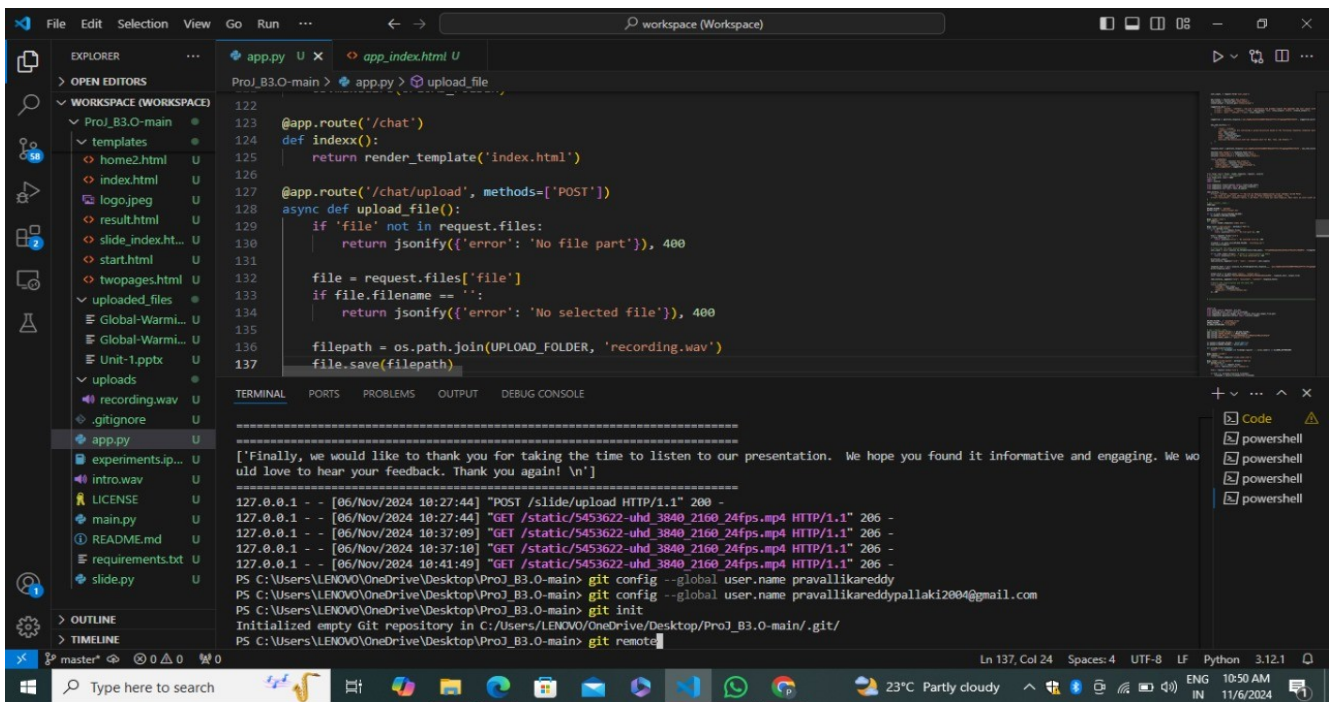
6.1 BLACK BOX TESTING:

Black box testing involves testing a system with no prior knowledge of its internal workings. A tester provides an input, and observes the output generated by the system under test. This makes it possible to identify how the system responds to the expected and a unexpected user actions , its response time, usability issues and the reliability issues. Black box testing is one of a powerful testing technique or architected, and expect to receive an appropriate response to their requests , a tester can simulate user activity and see if the system delivers on its promises . Along the way, a black box Test evaluates all relevant subsystems, including UI/UX, web server or application server, database,dependencies , and integrated systems.



6.2 WHITE BOX TESTING:

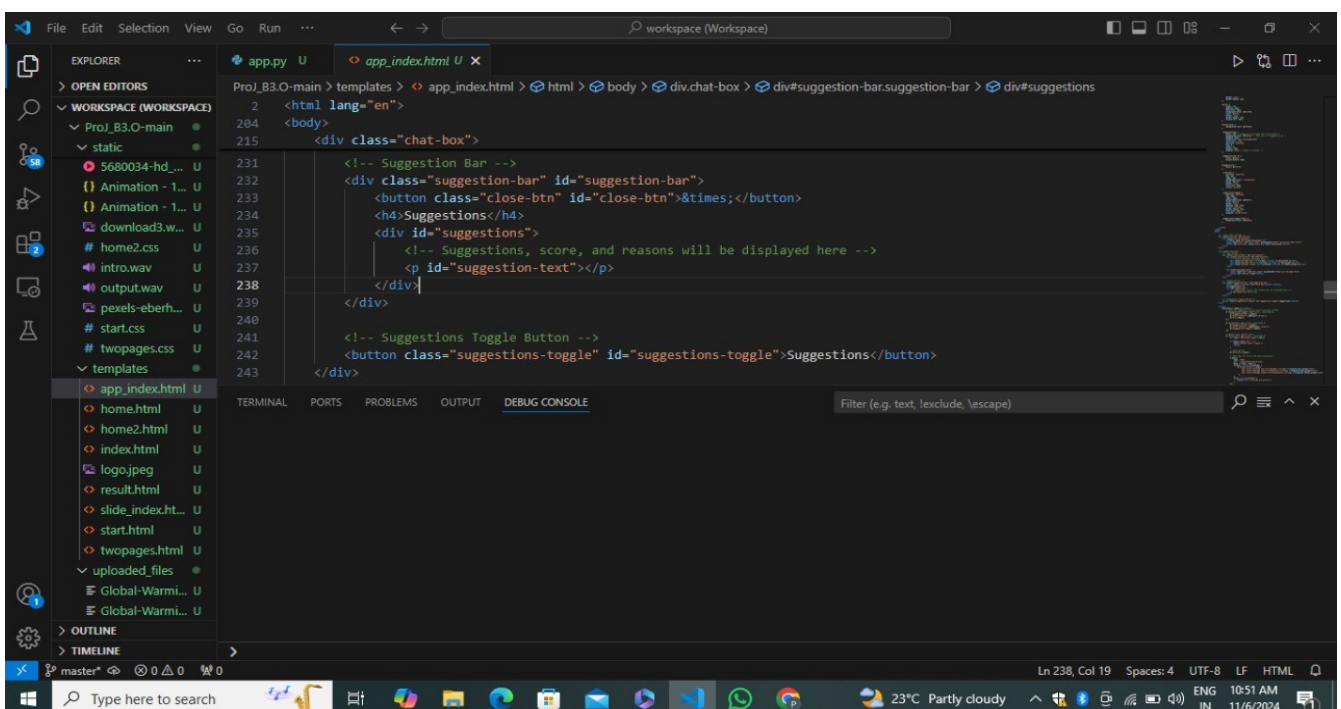
White box testing is the approach that allows testers to inspect and verify the inner workings of a software system — its code, infrastructure, and integrations with external systems. White box testing is an essential part of the automated build processes in the modern Continuous Integration/Continuous Delivery (CI/CD) development pipeline. White box testing is often referred to as of in the context of Static Application Security Testing (SAST), an approach that checks if the source code or binaries automatically and provides feedback on bugs and possible vulnerabilities.



The screenshot shows the VS Code editor with a Python file named `app.py` open. The code defines a Flask application with two routes: `/chat` and `/chat/upload`. The `/chat` route returns a render of `index.html`. The `/chat/upload` route handles file uploads, checking if a file is present and saving it to a specified folder. The terminal window shows the output of the application, including a message about feedback and a series of HTTP requests and responses.

```
122
123 @app.route('/chat')
124 def index():
125     return render_template('index.html')
126
127 @app.route('/chat/upload', methods=['POST'])
128 async def upload_file():
129     if 'file' not in request.files:
130         return jsonify({'error': 'No file part'}), 400
131
132     file = request.files['file']
133     if file.filename == '':
134         return jsonify({'error': 'No selected file'}), 400
135
136     filepath = os.path.join(UPLOAD_FOLDER, 'recording.wav')
137     file.save(filepath)
```

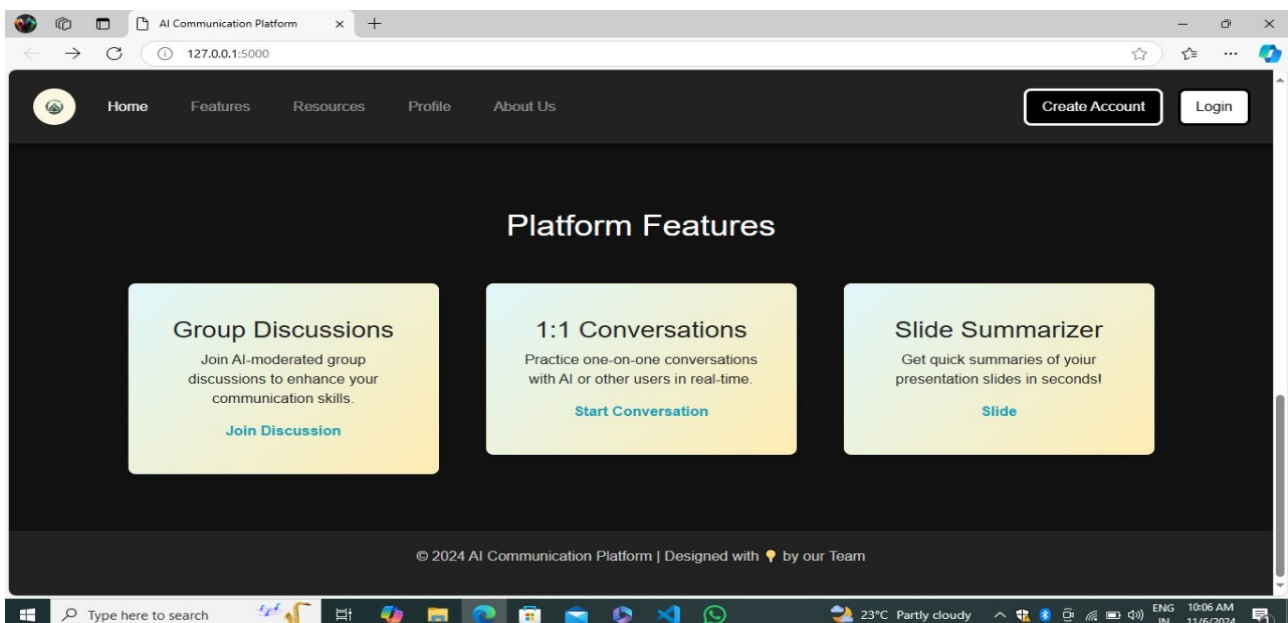
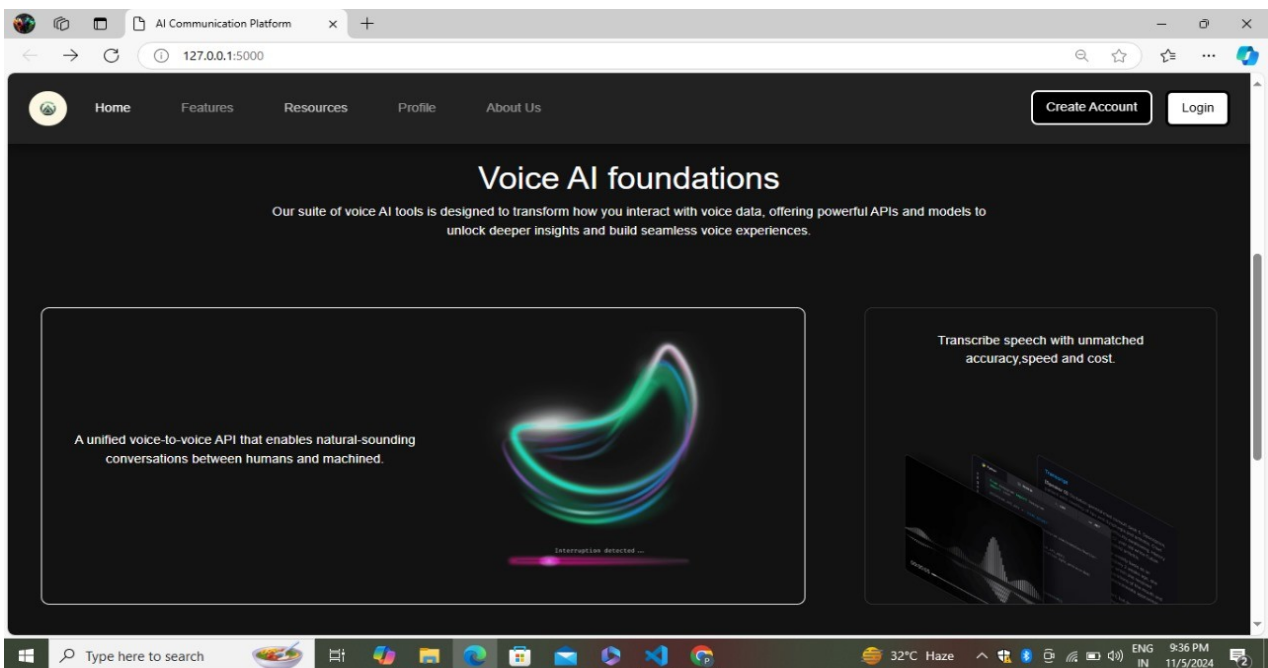
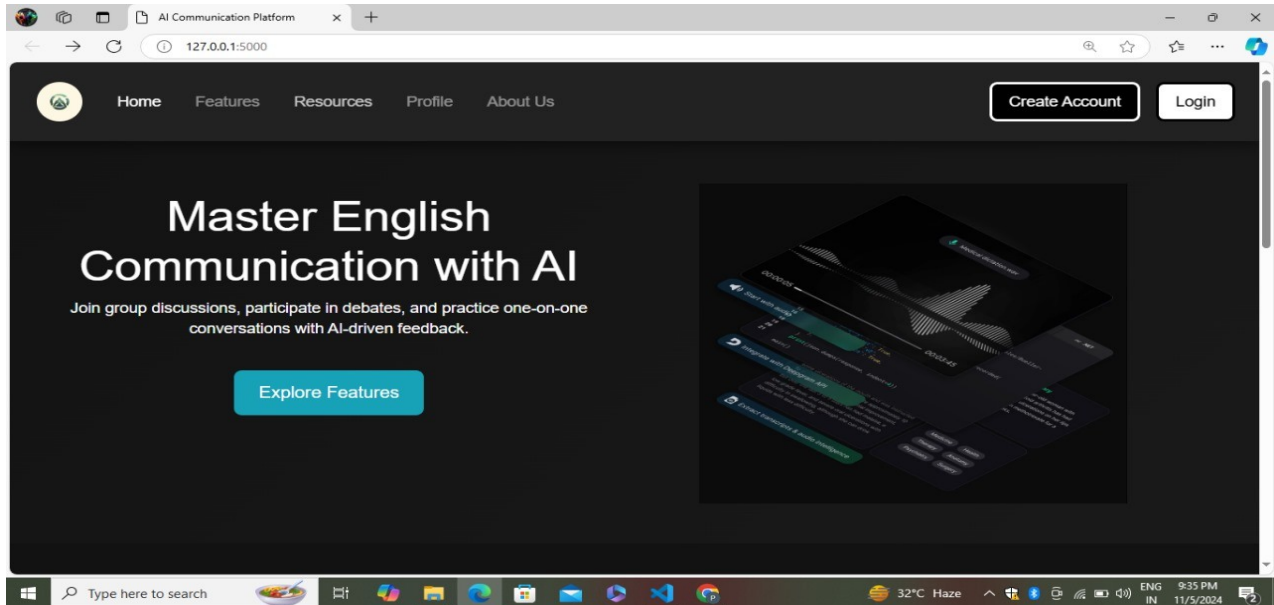
```
*****
[Finally, we would like to thank you for taking the time to listen to our presentation. We hope you found it informative and engaging. We wo
uld love to hear your feedback. Thank you again! \n']
*****
127.0.0.1 - - [06/Nov/2024 10:27:44] "POST /slide/upload HTTP/1.1" 200 -
127.0.0.1 - - [06/Nov/2024 10:27:44] "GET /static/5453622-uhd_3840_2160_24fps.mp4 HTTP/1.1" 206 -
127.0.0.1 - - [06/Nov/2024 10:37:09] "GET /static/5453622-uhd_3840_2160_24fps.mp4 HTTP/1.1" 206 -
127.0.0.1 - - [06/Nov/2024 10:37:10] "GET /static/5453622-uhd_3840_2160_24fps.mp4 HTTP/1.1" 206 -
127.0.0.1 - - [06/Nov/2024 10:41:49] "GET /static/5453622-uhd_3840_2160_24fps.mp4 HTTP/1.1" 206 -
PS C:\Users\LENOVO\OneDrive\Desktop\Proj_B3.O-main> git config --global user.name pravallikareddy
PS C:\Users\LENOVO\OneDrive\Desktop\Proj_B3.O-main> git config --global user.email pravallikaredypallaki2004@gmail.com
PS C:\Users\LENOVO\OneDrive\Desktop\Proj_B3.O-main> git init
Initialized empty Git repository in C:\Users\LENOVO\OneDrive\Desktop\Proj_B3.O-main\.git\
PS C:\Users\LENOVO\OneDrive\Desktop\Proj_B3.O-main> git remote
```

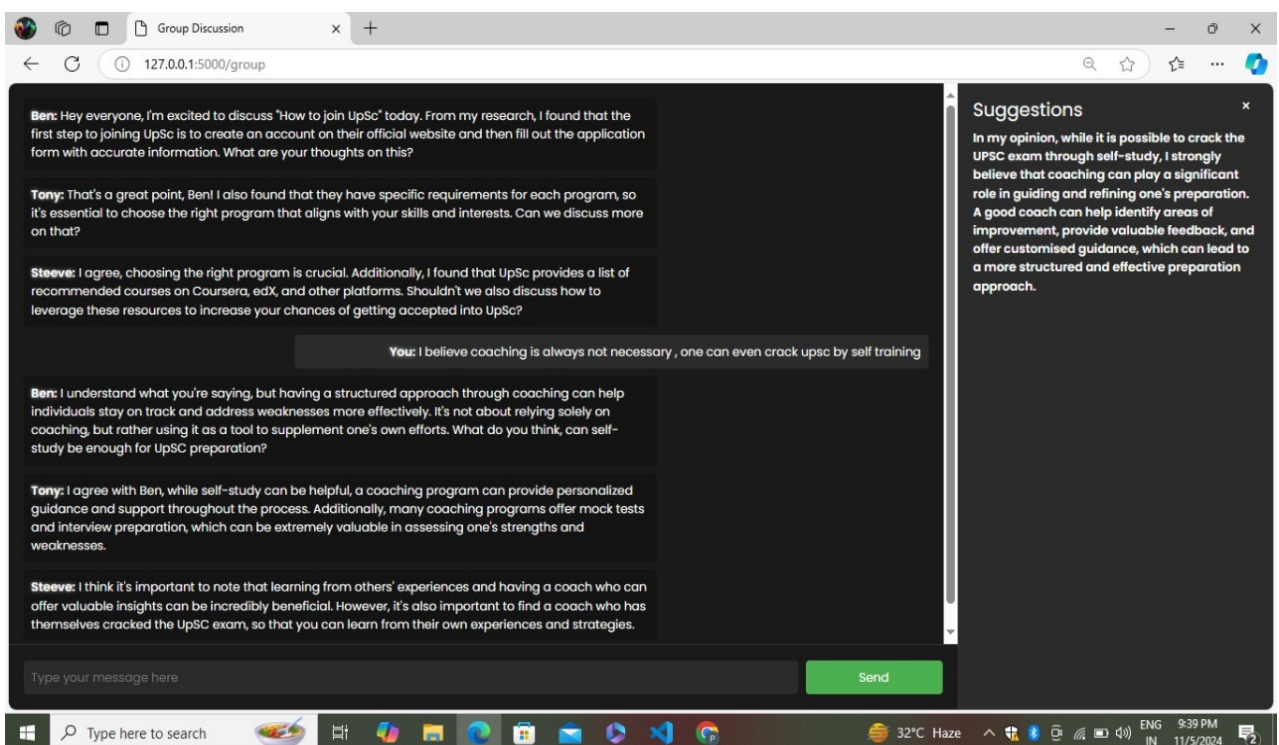
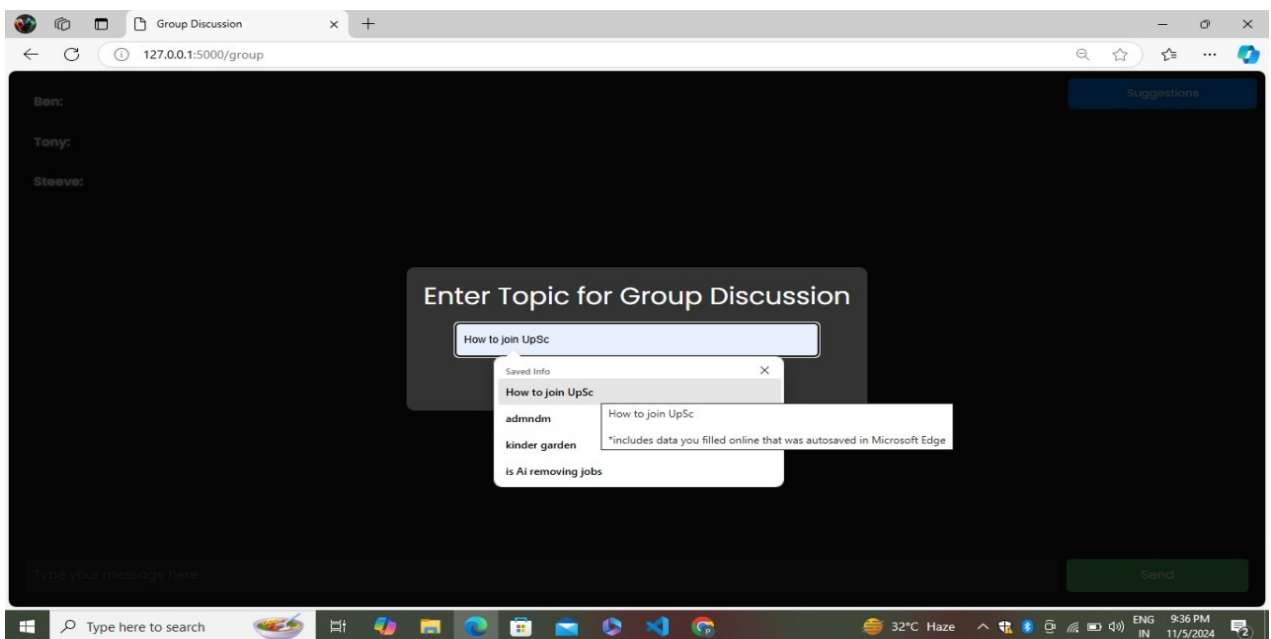
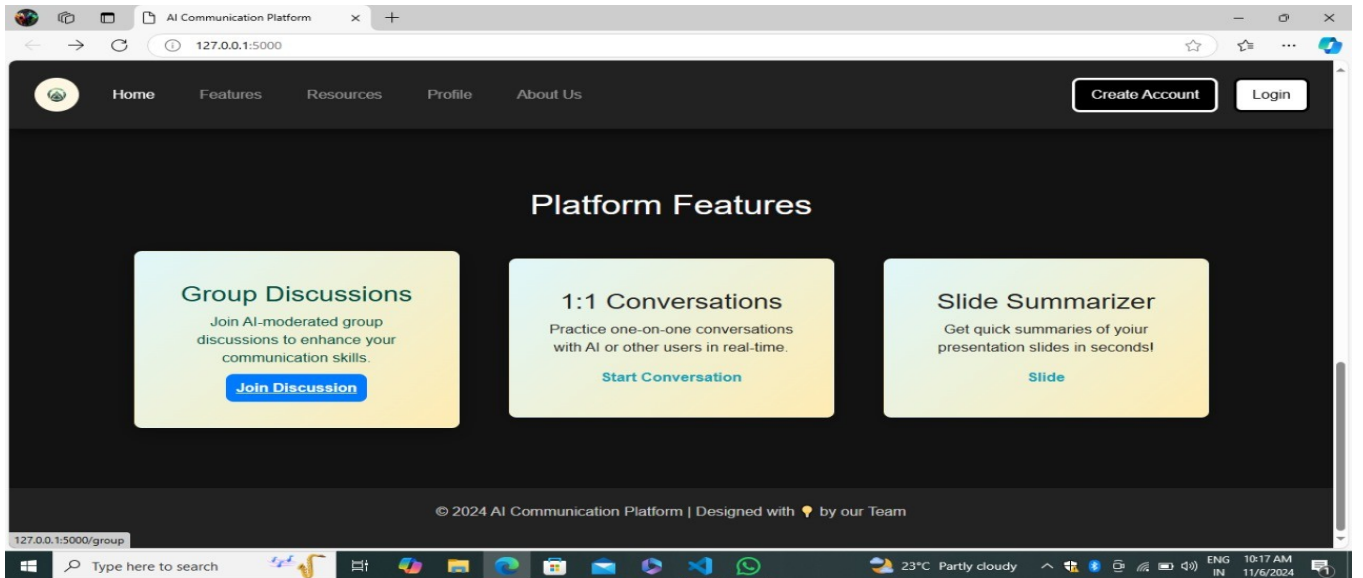


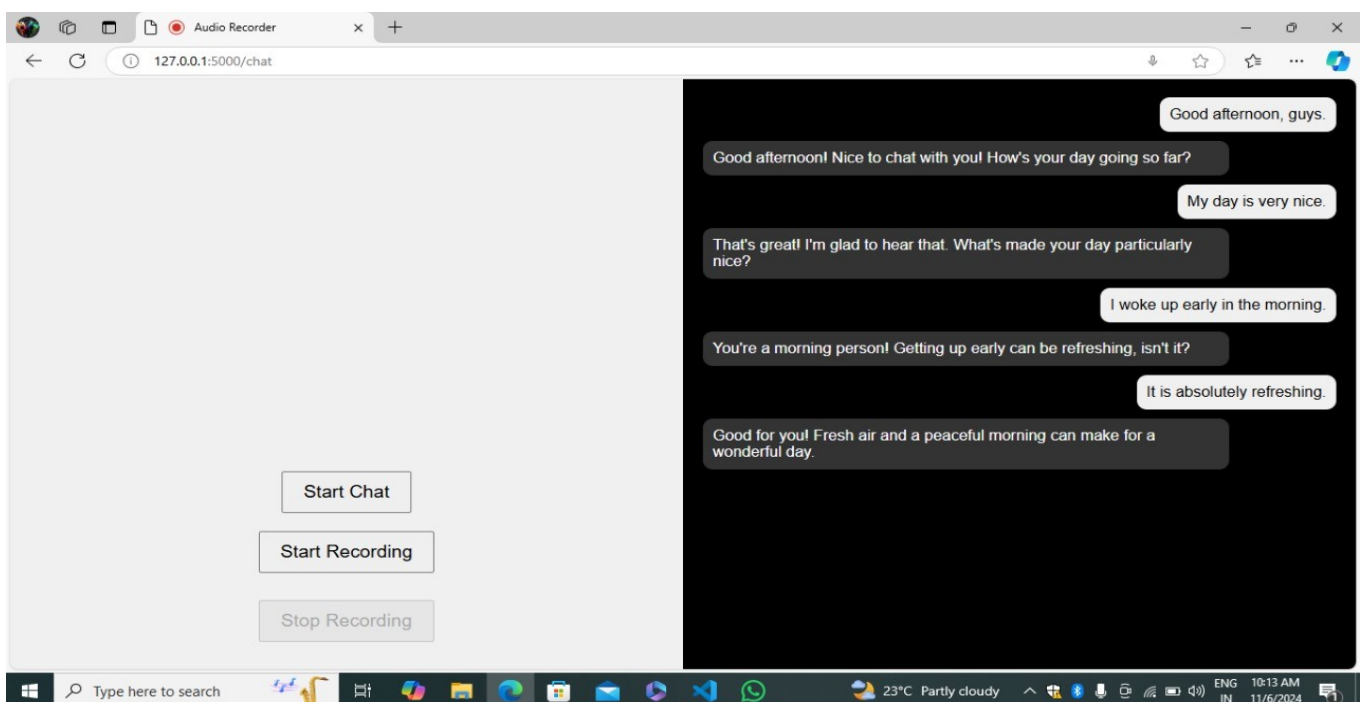
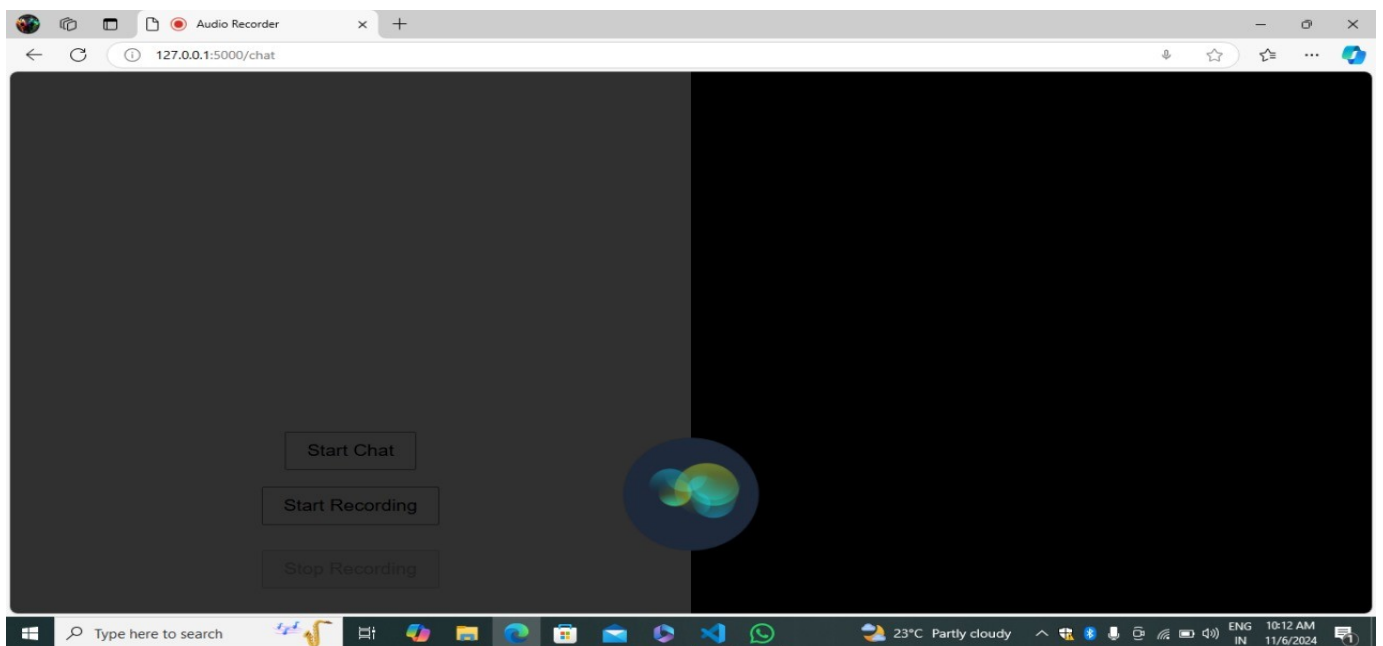
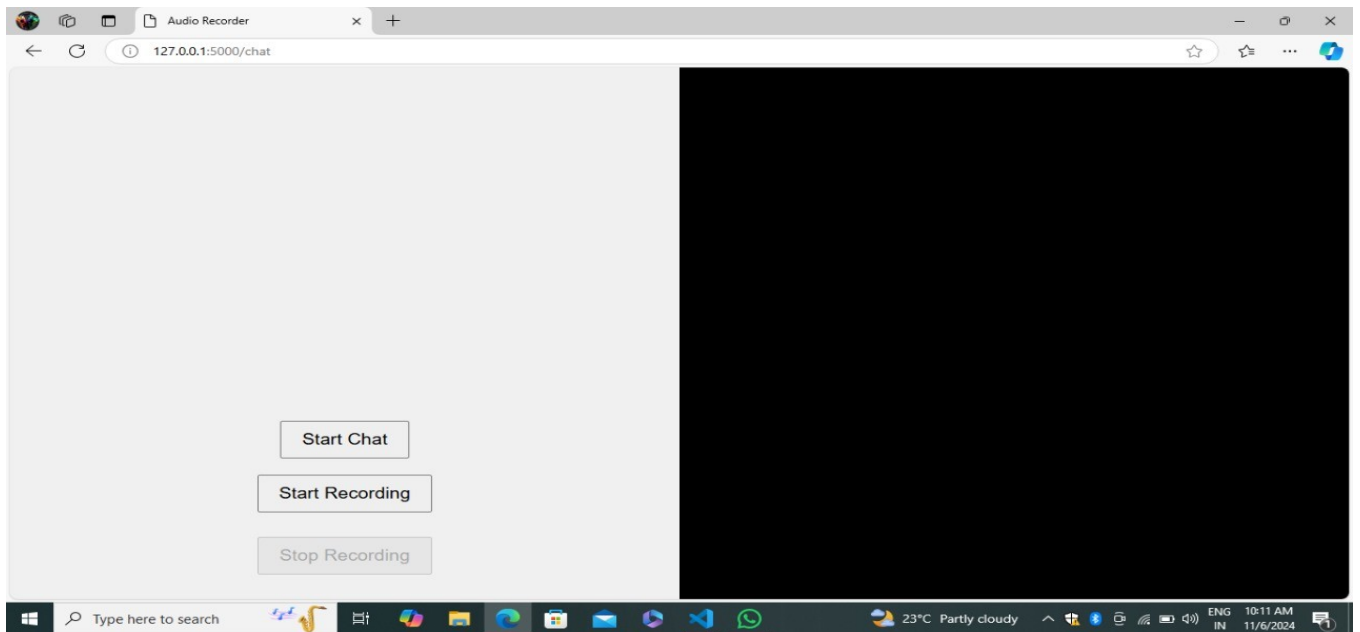
The screenshot shows the VS Code editor with an HTML file named `app_index.html` open. The code defines the structure of a chat interface, including a suggestion bar and a suggestions toggle button. The terminal window shows the output of the application, including a message about feedback and a series of HTTP requests and responses.

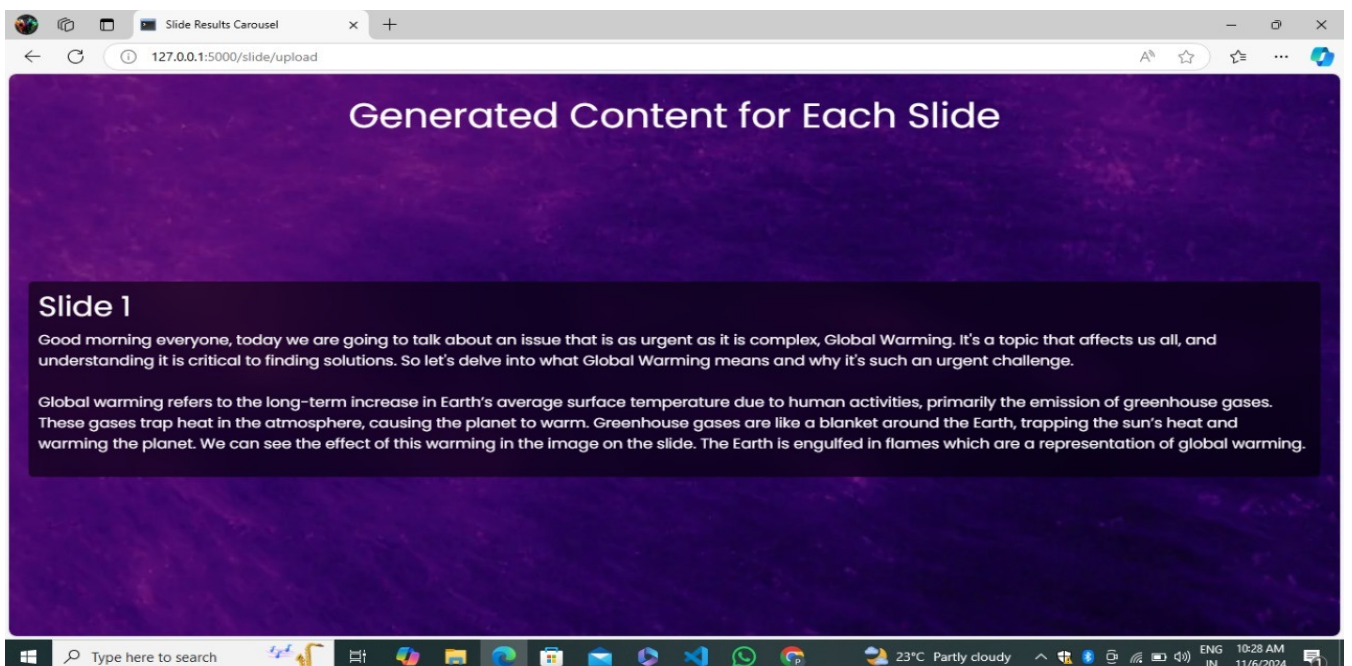
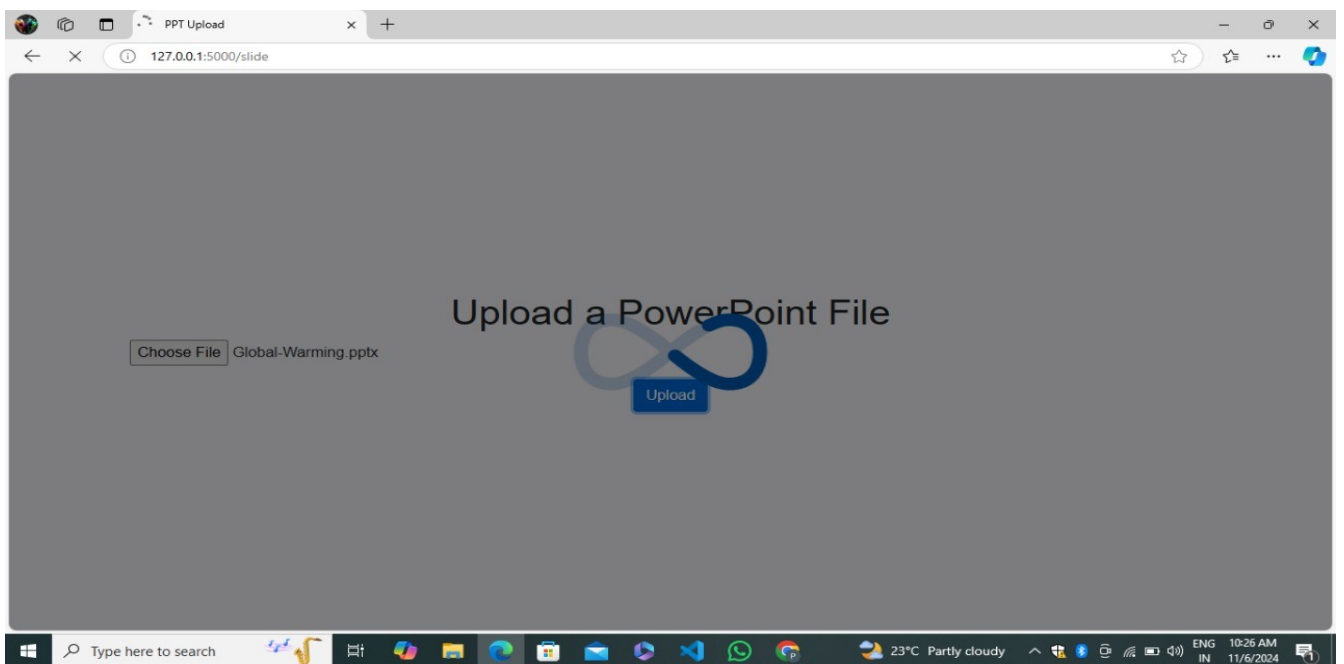
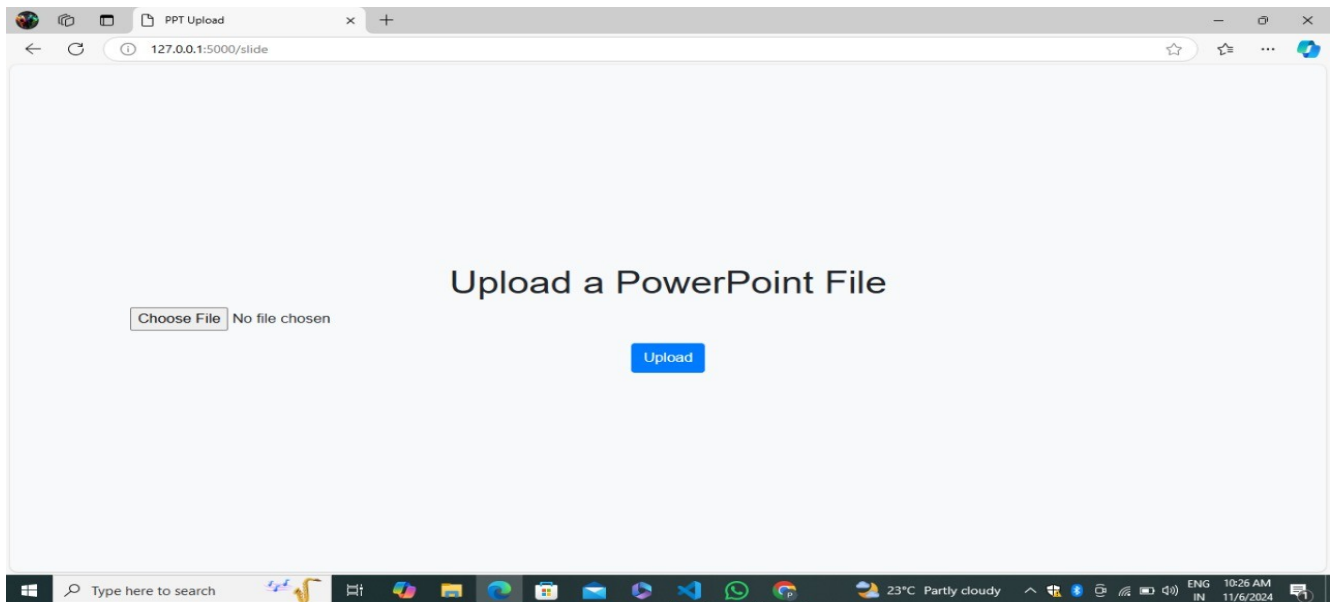
```
2 <html lang="en">
204 <body>
215 <div class="chat-box">
231 <!-- Suggestion Bar -->
232 <div class="suggestion-bar" id="suggestion-bar">
233 <button class="close-btn" id="close-btn">&times;</button>
234 <h4>Suggestions</h4>
235 <div id="suggestions">
236 <!-- Suggestions, score, and reasons will be displayed here -->
237 <p id="suggestion-text"></p>
238 </div>
239 </div>
240
241 <!-- Suggestions Toggle Button -->
242 <button class="suggestions-toggle" id="suggestions-toggle">Suggestions</button>
243 </div>
```

7.OUTPUT SCREENSHOTS









8.CONCLUSION

In conclusion, our AI-driven English communication platform provides an intuitive and engaging tool for improving English proficiency. By integrating group discussions, one-on-one conversations with AI, and slide summarization, we offer users a dynamic experience that adapts to diverse learning needs. Each feature includes interactive functionalities like real-time voice-to-text conversion, AI responses, and summarized insights from presentations, allowing users to practice and refine their skills. With a user-friendly interface and insightful visualizations, the platform enhances user engagement and learning efficiency. As we continue to improve and optimize the platform, we envision expanding its capabilities to further support effective English communication for users worldwide.

9.FUTURE ENHANCEMENT

In the future, we plan to enhance our AI-driven English communication platform by introducing user login and account features, allowing each user to create a personalized profile. This will enable users to track their progress over time, save conversation histories, and receive tailored feedback based on their previous interactions. Additionally, we aim to expand the platform's capabilities by incorporating more advanced AI models for real-time grammar correction and vocabulary suggestions during conversations. Another planned enhancement is to introduce multilingual support, which will allow non-native English speakers to practice English while having prompts and instructions available in their native language.

These future enhancements will provide users with a more personalized, effective, and engaging learning experience, making the platform a comprehensive tool for continuous improvement in English communication skills.

10.BIBILOGRAPHY

For successfully completing our project, We have taken help from the following website's links:

Websites:

These are the websites that we are referred while doing this project.

<https://deepgram.com/>

<https://www.fluentu.com/blog/english/learn-english-magazine/>

<https://github.com/DogukanOzyurt/English-Learning-App>

<https://flask.palletsprojects.com/en/stable/>

<https://groq.com/>